

CoW

Composable Poller

ENGAGEMENT	SECURITY REVIEW
REVISION	1.1
DELIVERED	2026-09-09
LEAD AUDITOR	Michal Převrátíl

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1. Document Revisions

1.0	Final Report	September 4, 2026
1.1	Fix Review	September 9, 2026

2. Overview

This document presents the findings of our review.

2.1. ack3

[ack3](#) is an in-house team of security researchers performing audits focused on manual code reviews with extensive fuzz testing and AI-assisted workflows for Ethereum and Solana. The ack3 team is trusted by top-tier organizations in web3, including Lido, Safe, and Axelar.

We develop the open-source security and developer tooling [Wake](#), which powers the fuzzing, static analysis, and graph-based review on our audits and helps auditors discover hardly recognizable edge-case vulnerabilities.

Our team teaches blockchain security at the [Czech Technical University in Prague](#), led by founder Josef Gattermayer, Ph.D.

For ack3, security is a long-term partnership: we work alongside our clients from design through post-deployment.

2.2. Review Methodology

We begin each review by mapping the codebase and the interactions between contracts and external components using [Wake](#)'s graph tooling. We work with the provided documentation and materials to discover discrepancies between the expected behavior and the actual implementation. Where applicable, we build a mathematical model of the protocol to cross-compare with the code and identify subtle bugs and inconsistencies.

At the start of the review, we launch an automated [ack3](#) AI scan pinned to the scoped commits, running in parallel with the manual review. The scan provides additional coverage and new leads for the team to analyze; each scan finding is manually reviewed by the team.

During the review, we apply AI-assisted workflows and skills that support codebase onboarding and navigation, combined with Wake's static analyzer and language server for context-aware code analysis.

We launch AI agents to build on our vulnerability hypotheses and expand the attack surface with new escalation paths. All findings are manually reviewed and validated by the team. Where applicable, we confirm findings and verify the exact behavior of the code with unit tests and fuzzing. Where the engagement includes a fuzzing campaign, it is manually guided by an auditor; the fuzz tests themselves are AI-generated.

Once sufficient context is gathered, we search for deeper, chained issues based on the attack vectors discovered during review. This includes combining smaller weaknesses, testing trust-boundary assumptions, and evaluating whether unexpected behavior can lead to practical impact.

The validated results are documented in the report: findings with severity, affected components, impact, and recommended remediation.

2.3. Finding Classification

The Severity of each finding combines two sub-ratings, Impact and Likelihood, and ranges from Info to Critical.

If we find a scenario in which an issue is exploitable, it is assigned an impact rating of High, Medium, or Low, based on its consequences for the system. If we find no way to exploit it, or the issue is only exploitable given a change in configuration (system settings or parameters, such as deployment scripts, compiler configurations, or multi-signature wallets for owners) or in the codebase, it is assigned Warning or Info.

Low to High impact issues also have a Likelihood, which measures the probability of exploitation at runtime.

Severity

		LIKELIHOOD			
		HIGH	MEDIUM	LOW	N/A
IMPACT	HIGH	Critical	High	Medium	-
	MEDIUM	High	Medium	Low	-
	LOW	Medium	Low	Low	-
	WARNING	-	-	-	Warning
	INFO	-	-	-	Info

Table 1. Severity of findings

Impact

- **High** - Code that activates the issue leads to undefined or catastrophic consequences for the system.
- **Medium** - Code that activates the issue results in serious consequences.
- **Low** - Code that activates the issue has consequences that are recoverable or do not jeopardize the system's normal operation.
- **Warning** - The issue cannot be exploited given the current code or configuration, but could become a security vulnerability if either changed. If we have not found a way to exploit the issue given the time constraints, it may be marked as Warning or higher, based on our best estimate of whether it is currently exploitable.
- **Info** - The issue is on the borderline between code quality and security. Examples include insufficient logging for critical operations, or issues that would become security-related if code or configuration changed.

Likelihood

- **High** - The issue is exploitable by virtually anyone under any circumstance.
- **Medium** - Exploiting the issue requires non-trivial preconditions.
- **Low** - Exploiting the issue requires strict preconditions.

2.4. Review Team

For a revision’s authors, see the “Revision team” section in that revision’s chapter.

NAME	POSITION
Michal Převrátíl	Lead Auditor
Naoki Yoshida	Auditor
Josef Gattermayer, Ph.D.	Audit Supervisor

2.5. Disclaimer

We have made our best effort to find all vulnerabilities in the system; however, our findings should not be considered a complete list of issues. Statements in this document should not be interpreted as investment or legal advice, nor should the authors be held accountable for decisions based on them.

3. Executive Summary

CoW is a DEX aggregation protocol that settles user orders in batch auctions executed by competing solvers, with the ComposableCoW framework extending it to conditional orders whose parts become tradeable over time. Composable Poller funds these conditional orders just in time, transferring the sell amount from the funder to the order owner only once a part is ready to trade.

Revision 1.0

CoW engaged [ack3](#) to perform a security review of CoW Composable Poller with a total time donation of 2 engineering days over a period between August 31 and September 4, 2026, with Michal Převrátíl as the lead reviewer.

The review was performed on the commit [3ff3cf5^{\[1\]}](#) in the [composable-cow](#) repository, and the scope was the `src/types/ComposableCowPoller.sol` file.

We began our review by preparing a differential, manually guided fuzz test in the [Wake](#) framework, modeling the expected behavior of the contract and comparing it to the actual behavior. We then proceeded with a deep manual review of the code. At the end of the review, we launched an automated AI scan and reviewed all generated vulnerability hypotheses.

During the review, we focused in particular on the following aspects:

- the signature verification captures all required fields;
- the register and revoke actions are properly validated for authorization;
- the contract is protected against replay attacks;
- the contract correctly integrates with the other CoW contracts;
- [EIP-712](#) is implemented consistently with the specification; and
- the documentation is accurate and users are informed about the trust

boundaries.

Our review resulted in a single informational finding, [1](#). The finding describes inaccurate documentation of the trust boundaries between the ComposableCowPoller contract in scope and the rest of the CoW system. The code quality exceptionally good.

[ack3](#) recommends CoW to:

- clearly document which CoW handlers are supported by the poller;
- clearly document the trust assumptions for the handler; and
- consider whitelisting only the first-party supported handlers.

See [Report Revision 1.0](#) for the system overview and trust model.

Revision 1.1

CoW engaged [ack3](#) to perform a fix review of the finding identified in the previous revision.

The review was performed on September 9, 2026, on commit [665ee6d](#)^[2].

The previously reported finding [1](#) was confirmed to be fixed. No new findings were identified during the review.

[1] full commit hash: 3ff3cf519f2b166713b8c5ff309bde14d6f0dff4, link to [commit](#)

[2] full commit hash: 665ee6d38176c509da5356c3d74285b3f8f2b99d, link to [commit](#)

4. Findings Summary



Figure 1. Findings by Severity

FINDING TITLE	SEVERITY	REVISION	STATUS
1: Inaccurate poller documentation	INFO	1.0	Fixed

Table 2. Table of Findings

Report Revision 1.0

Revision Team

NAME	POSITION
Michal Převrátíl	Lead Auditor
Naoki Yoshida	Auditor
Josef Gattermayer, Ph.D.	Audit Supervisor

System Overview

`ComposableCowPoller` provides just-in-time funding for `ComposableCoW` conditional orders, keeping the sell amount with the funder until each part is ready to trade. A funder registers a schedule that pins the conditional-order handler, the order owner, a salt, and the order's `staticInput`, and whose key is derived from the funder, handler, owner, and salt. Schedules are registered and revoked by the funder directly, by an [EIP-712](#) signature that any relayer may submit before its deadline, or by the funder's own `CowShed` proxy resolved through the factory pinned at deployment. Registrations and delegated revocations carry an authorization epoch that must match storage; revocation clears the schedule and increments the epoch, invalidating earlier authorizations while leaving the key reusable. The permissionless `pollFunds`, typically called from a `CoW` Protocol pre-hook, re-derives the order's params hash, requires it to be authorized in `ComposableCoW.singleOrders`, and reads the current order from the handler. It then transfers the order's `sellAmount` from the funder to the owner through the funder's allowance and records the order digest, so each digest is funded only once.

Trust Model

The poller distinguishes two roles: the funder, which supplies the sell tokens, and the owner, which holds the conditional order in `ComposableCoW` and receives the pulled funds. They may be the same address. The contract holds no funds and grants itself no privileges; every transfer draws on the token allowance the funder grants it.

The funder fully trusts the owner. `pollFunds` moves the sell amount to the owner, and the poller places no constraint on those funds afterwards. The owner also authorizes and removes its own conditional orders in `ComposableCoW`, so removing an order pauses funding only while the same `paramsHash` stays unauthorized: reauthorizing it resumes funding without new funder consent, as long as the schedule and the allowance remain. The funder's exposure is therefore bounded by the granted allowance rather than by the state of the conditional order. Revoking the schedule stops further funding under it, but leaves the allowance in place for any schedule registered later.

The funder is responsible for selecting a compatible handler and configuring it in a compatible way. `pollFunds` takes the sell token, the sell amount, and the time validity from `getTradeableOrder`; it does not pin the token at registration, cap the amount, or check `validTo` after the handler returns. Replay protection is keyed to the exact order digest, so funding a part once holds only for handlers whose output is deterministic per part. A handler whose output depends on mutable state, such as an oracle price or a balance, yields a new digest whenever that state changes, and each digest is funded once. The `staticInput` stored in the schedule is neither part of the schedule key nor validated at registration; a value that does not match the authorized order makes `pollFunds` revert `OrderNotLive`. These boundaries are documented inaccurately, which is the subject of [1](#).

`COMPOSABLE_COW` and `COW_SHED_FACTORY` are pinned at deployment and trusted

thereafter. The factory must derive proxies from the owner alone, because a call from `proxyOf(funder)` is what proves the funder authorized a `FromShed` action, and it must deploy sheds that forward [ERC-1271](#) signatures, because a shed without forwarding can be funded but cannot settle. `pollFunds` is permissionless: any caller can trigger the transfer while the handler's window is open, so the funder does not control its timing.

Findings

This section lists the findings from this revision; for all findings, see the [Findings Summary](#).

I1: Inaccurate poller documentation

Impact:	Info	Likelihood:	N/A
Type:	Code quality	Discoverer:	Human
Target:	README.md, ComposableCowPoller.sol		

Description

The README and NatSpec document stronger or stale guarantees than ComposableCowPoller enforces. The implementation is internally consistent, but the current wording can mislead funders and integrators about the boundary provided by a registered schedule and a standing token allowance.

The README states that pollFunds verifies that the order is active, transfers only the current part, and funds each part once.

Listing 1. Excerpt from [README](#)

```
121 `ComposableCowPoller` enables an EOA to create a ComposableCoW conditional
    order, such as a TWAP, without moving the full sell amount into the
    order's smart account up front. The funds remain in the EOA until each
    part is ready to trade. A pre-hook then calls `pollFunds`, which verifies
    that the order is still active and transfers only that part's sell amount
    to the smart account. Each part can be funded only once.
```

pollFunds checks only that the registered conditional order hash is currently authorized in ComposableCoW.singleOrders, then asks the handler for the order. Replay protection is keyed to exact orderDigest reuse, and the transferred token and amount are taken from the returned order. For handlers whose output can change with mutable state, the accurate guarantee is per digest, not per conceptual part.

Listing 2. Excerpt from [ComposableCowPoller.pollFunds](#)

```
440 // The order must still be authorised; `remove` disables the poller.
```

```
441 if (!COMPOSABLE_COW.singleOrders(schedule.owner, paramsHash))
442     revert OrderNotLive();
```

Listing 3. Excerpt from [ComposableCowPoller.pollFunds](#)

```
445 GpV2Order.Data memory order = schedule.handler.getTradeableOrder(
446     schedule.owner,
447     address(this),
448     paramsHash,
449     schedule.staticInput,
450     bytes(""))
```

Listing 4. Excerpt from [ComposableCowPoller.pollFunds](#)

```
454 bytes32 orderDigest = GpV2Order.hash(
455     order,
456     COMPOSABLE_COW.domainSeparator()
457 );
458 if (funded[id][orderDigest]) return false;
459 funded[id][orderDigest] = true;
```

Listing 5. Excerpt from [ComposableCowPoller.pollFunds](#)

```
461 order.sellToken.safeTransferFrom(
462     schedule.funder,
463     schedule.owner,
464     order.sellAmount
```

The README also says removing the conditional order stops further funding. That is true only while the same paramsHash remains unauthorized; if the owner reauthorizes it while the poller schedule and token allowance remain, pollFunds can fund again without new funder consent.

Listing 6. Excerpt from [README](#)

```
127 Removing the conditional order from ComposableCoW stops further funding.
    Revoking its funding schedule does too, but does not remove the funder's
    token allowance; revoke that allowance separately when it is no longer
    needed.
```

ScheduleRegistered describes an update or replacement path, but `_register` rejects an occupied schedule key. Replacement requires revocation and a new registration.

Listing 7. Excerpt from [ComposableCowPoller](#)

```
113 /// @notice Emitted when a schedule is registered or updated.
114 /// @param id The deterministic key of the schedule.
115 /// @param owner The conditional-order owner and pull destination.
116 /// @param funder The token source that registered the schedule.
117 /// @param authEpoch The schedule's authorization epoch.
118 /// @param paramsHash The ComposableCow order key this schedule funds.
    `id` deliberately excludes
119 ///         `staticInput`, so re-registering the same funder, handler,
    owner, and salt replaces
120 ///         the stored schedule. Logging the hash names the order each
    registration points at,
```

Listing 8. Excerpt from [ComposableCowPoller.register](#)

```
263 if (schedules[id].funder != address(0)) revert AlreadyRegistered();
```

funded says replay history survives schedule updates, but active schedules are not updated. The history instead survives revocation and re-registration.

Listing 9. Excerpt from [ComposableCowPoller](#)

```
79 /// @dev `id => orderDigest => funded`. History survives schedule updates
    so an old order cannot be replayed.
```

OnlyFunder says it applies to registration, updates, and revocation, but the error is used only by direct registration.

Listing 10. Excerpt from [ComposableCowPoller](#)

```
82 /// @notice Thrown when someone other than the schedule funder registers,
    updates, or revokes a schedule.
83 error OnlyFunder();
```

Listing 11. Excerpt from [ComposableCowPoller.register](#)

```
195 if (msg.sender != schedule.funder) revert OnlyFunder();
```

The handler-window comment states a handler behavior as a poller invariant. pollFunds does not read validTo after getTradeableOrder returns, so the window holds only for handlers that revert outside it. The comment should present the window as a trust assumption on the handler to which pollFunds delegates time validity, not as a check the poller performs.

Listing 12. Excerpt from [ComposableCowPoller.pollFunds](#)

```
444 // The handler yields the current order and reverts outside its window.  
445 GPv2Order.Data memory order = schedule.handler.getTradeableOrder(
```

Recommendation

Update the README and NatSpec to match the implemented invariants. State that pollFunds is permissionless, checks the registered schedule and ComposableCoW.singleOrders, delegates token, amount, and time validity to the handler, records funding per orderDigest, and only pauses after conditional-order removal until the order hash is reauthorized unless the schedule is also revoked. Also update the stale comments for schedule replacement, replay history, OnlyFunder, and handler window checks.

Fix 1.1

The README and NatSpec comments were updated to better reflect the trust model and current implementation.

[Go back to Findings Summary](#)

Report Revision 1.1

Revision Team

Revision team is the same as in [Report Revision 1.0](#).

Appendix A: How to cite

Cite this document as:

[ack3](#), Security Review | CoW: Composable Poller, September 9, 2026.



Thank you.

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